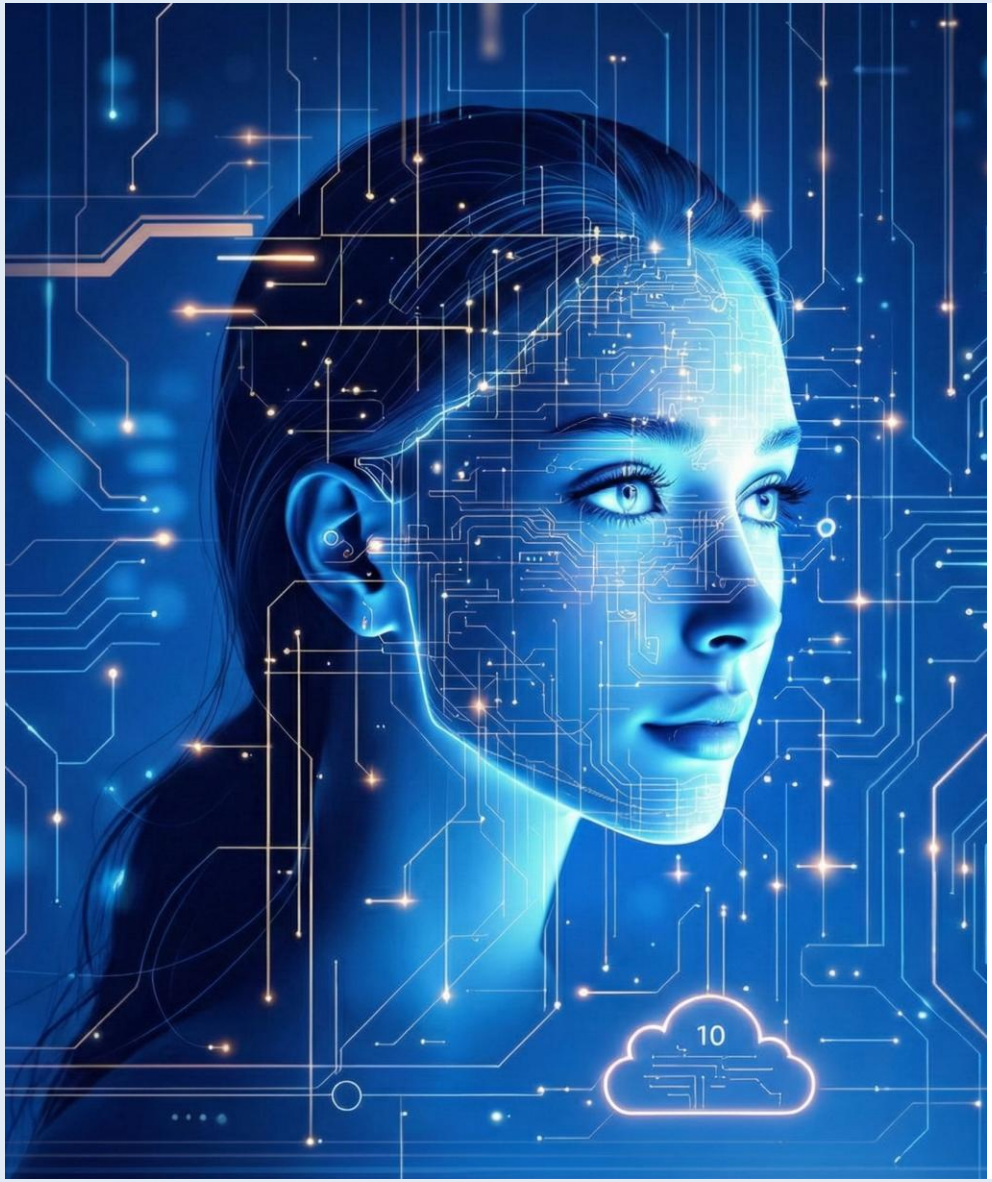

FIVUCSAS

FACE AND IDENTITY VERIFICATION USING CLOUD-BASED SAAS MODELS

CSE4297 ENGINEERING PROJECT I

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OUTLINE

WHY

1. The Problem & Motivation
2. Related Work & Market Gap
3. Scope

HOW

4. Methodology and Technical Approach
5. The Biometric Puzzle
6. Card & NFC Verification
7. Machine Learning Pipeline

WHAT

8. Tasks Accomplished
9. Difficulties Encountered
10. Tasks to be Completed

THE PROBLEM & MOTIVATION

The Threat is Real

$

$23B

Identity Fraud Losses (U.S.) [1]

+ 1000 %

Deepfake Surge (2022-2023)[2]

Fig. 1: Face-to-Embedding Transformation Concept

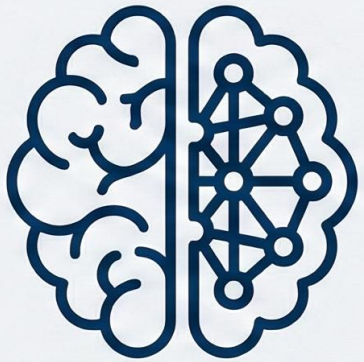
?

3 in 4

Cannot Detect Deepfakes[3]

Sources: Javelin/AARP 2024; Sumsub 2024; iProov 2025

RELATED WORK



Deep Face Recognition

DeepFace, FaceNet, ArcFace
Embedding-based verification
Accuracy-focused

Liveness Detection

Passive + Active methods
Anti-spoofing
Isolated solutions

Integration Gap

No multi-tenant support
No cloud-native deployment
No end-to-end system

Models → Detection → Gap → **Our Solution** (FIVUCSAS)

RELATED WORK COMPARISON

	Azure Face Liveness	AWS Rekognition	Sodec Technologies	FaceAuth Vision	BioGATE Pass	FIVUCSAS
Open Source	✗	✗	✗	✓	✗	✓
Liveness Detection	✓	✓	✗	✗	✓	✓
Multi-Tenant SaaS	✓	✓	✗	✗	✗	✓
Multi-Platform	✓	✓	✓	✓	✗	✓
PAD Compliance	✓	✓	✗	✗	✓	✓

The Gap

Most solutions lack physical document verification (NFC) and rely on singular, predictable liveness checks.

Our Advantage

We bridge the physical-digital gap and combine active and passive liveness into an unpredictable "Biometric Puzzle."

SCOPE

IN SCOPE
OUT OF SCOPE
CONSTRAINTS
100kb
Image Quality
>100kb
Latency
<200ms
ICAO Doc
9303
Six core deliverables defined
Hardware Manufacturing
Strict performance and standards

METHODOLOGY AND TECHNICAL APPROACH

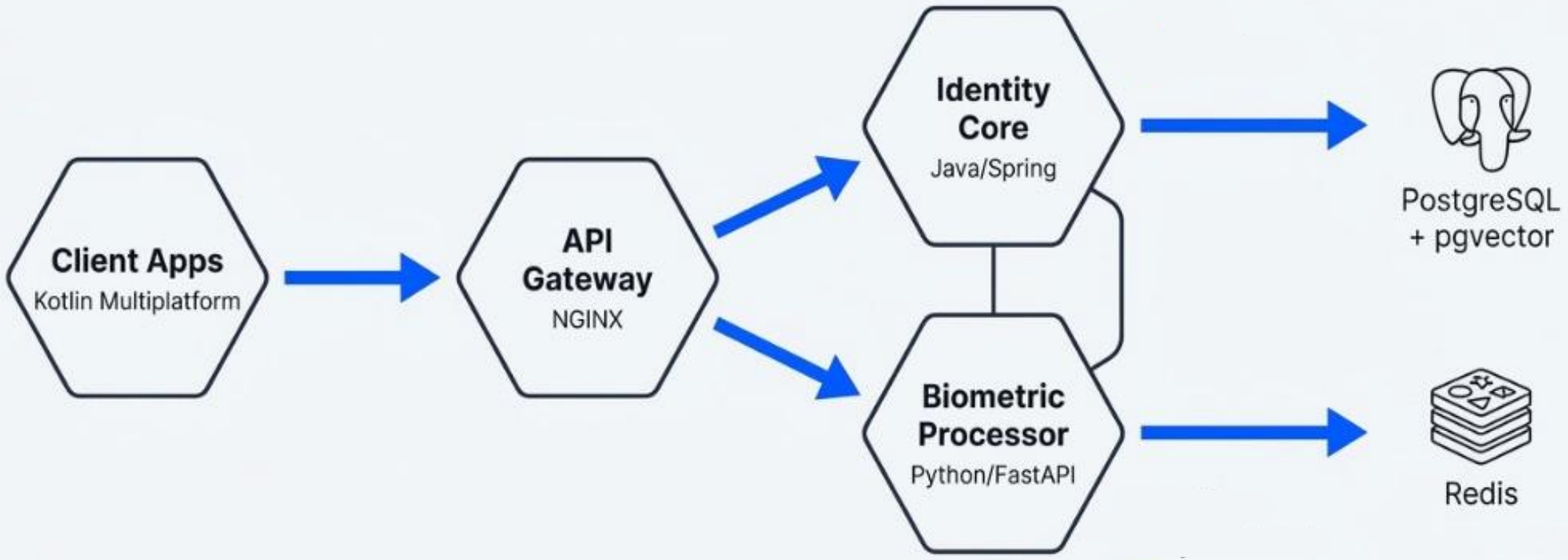


Fig. 2: Microservices Architecture Overview

THE BIOMETRIC PUZZLE

OUR INNOVATION

- Hybrid Liveness Detection
- Anti-Deepfake Protection

1. Server generates random challenge
   → "Look Direct", "Close Eyes"
   "Smile Wide", "Turn Head Right"

2. Client captures video stream
   → 468 facial landmarks tracked
   → EAR calculation: < 0.2 = Close Detected
   → MAR calculation: > 0.6 = Mouth Open

3. Parallel passive texture analysis

Fig. 3: Multi-Action Liveness Detection with Real-Time Facial Landmark Tracking

CARD DETECTION & NFC VERIFICATION

Fig. 4: Visual Document Detection Example

Fig. 5: NFC Chip Verification Steps

ML PIPELINE & VECTOR SEARCH

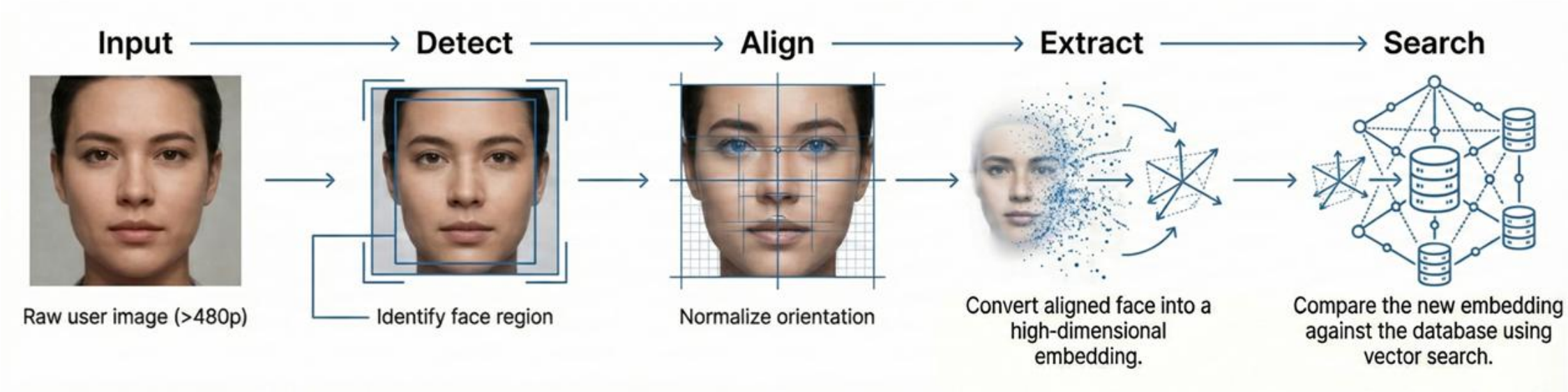


Fig. 6: Face Recognition Pipeline

BIOMETRIC ANALYSIS - 1

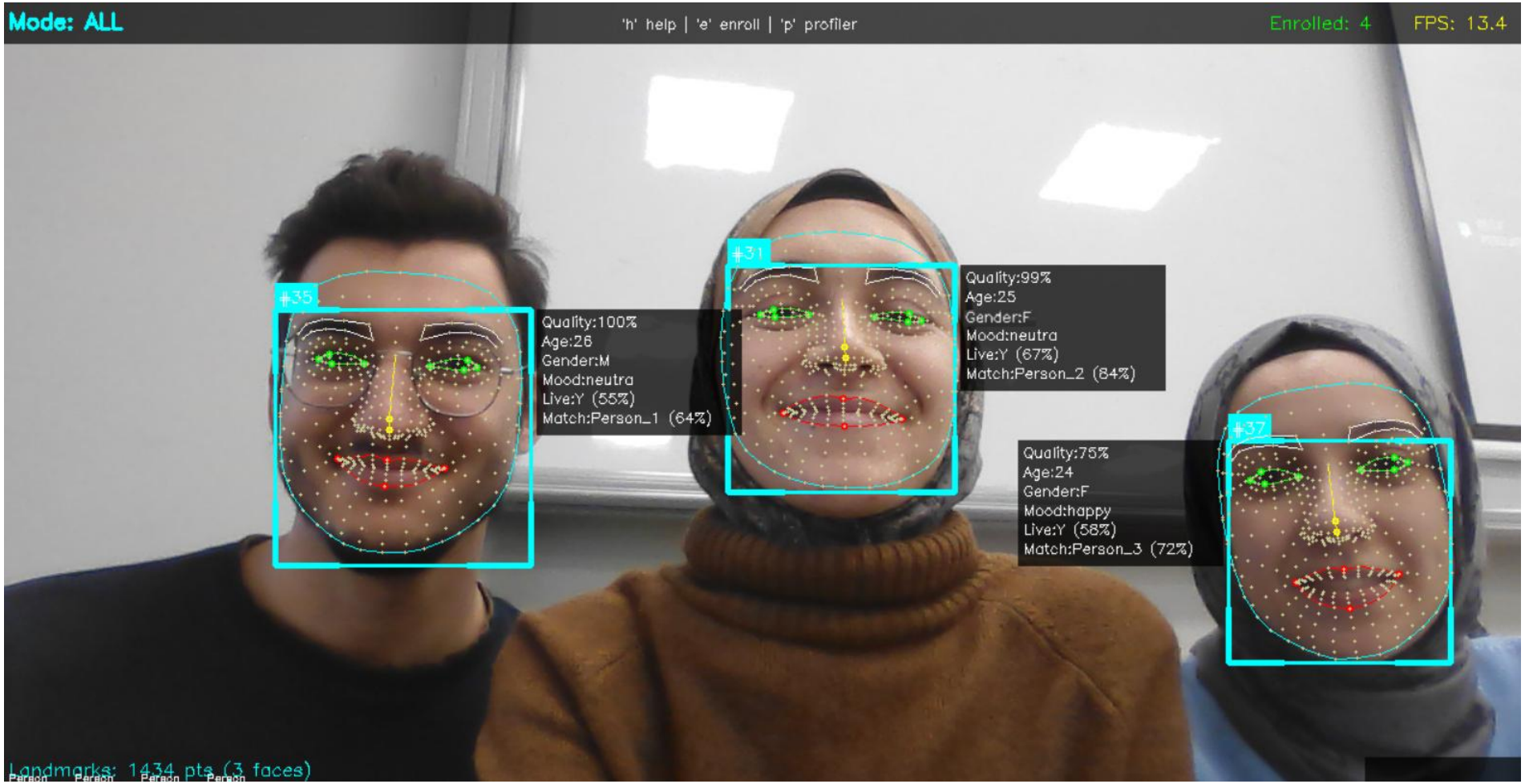


Fig. 7: Multi-Subject Biometric Analysis with Real-Time Attribute Extraction

BIOMETRIC ANALYSIS - 2

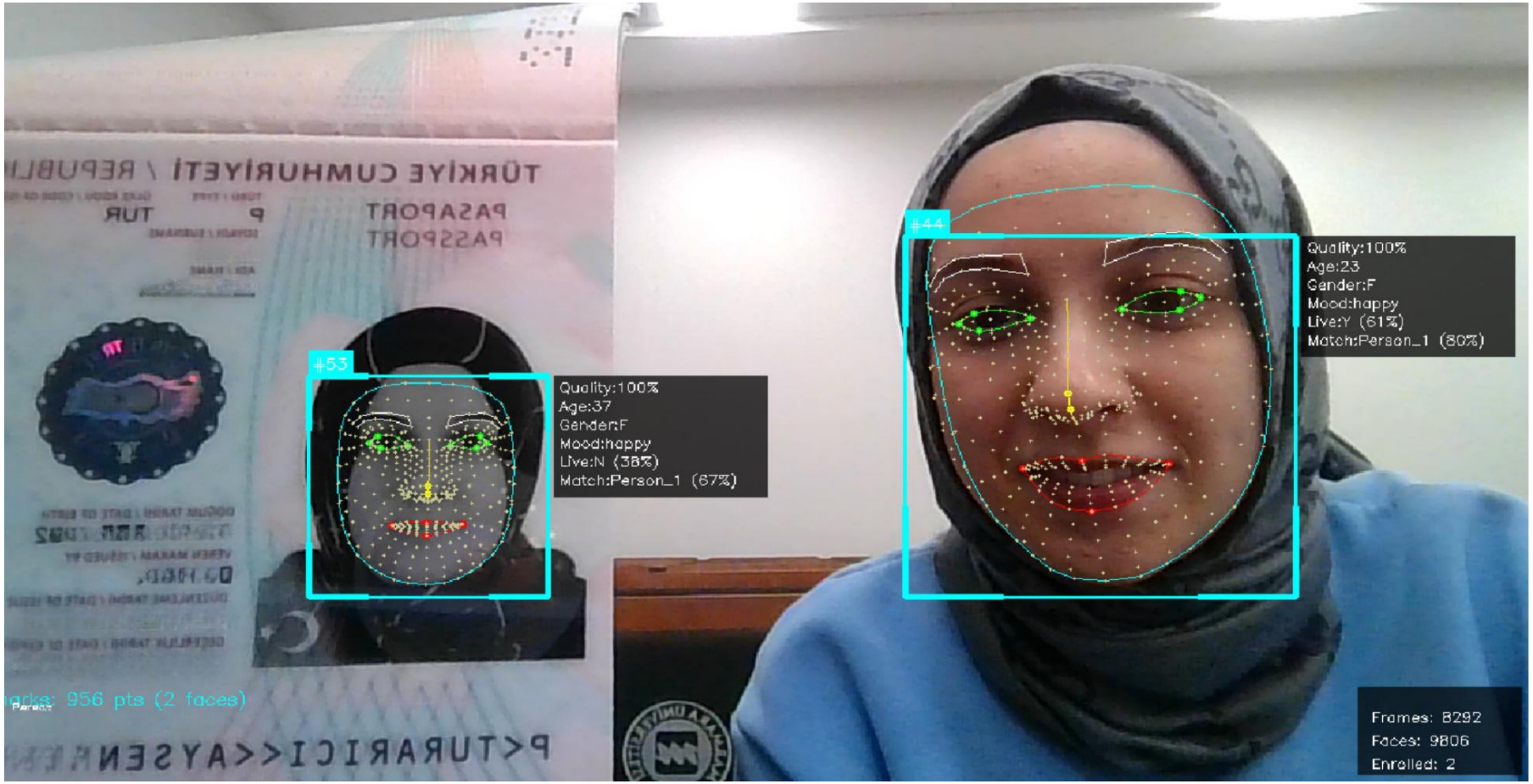


Fig. 8: Biometric Identity Verification — Passport-to-Live Face Comparison

BIOMETRIC ANALYSIS - 3

Fig. 9: Real-Time Travel Passport Detection Using Custom-Trained ML Model

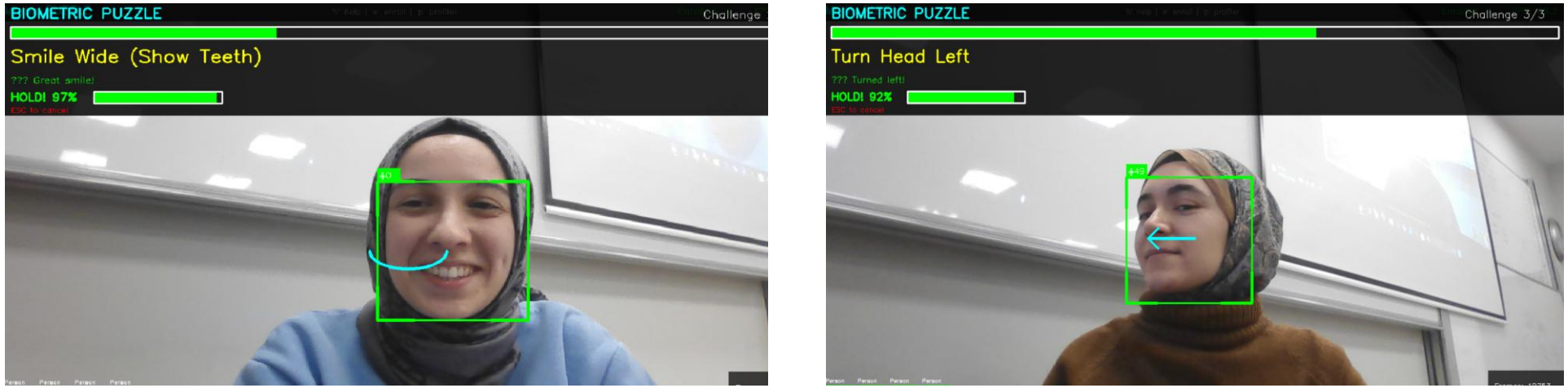
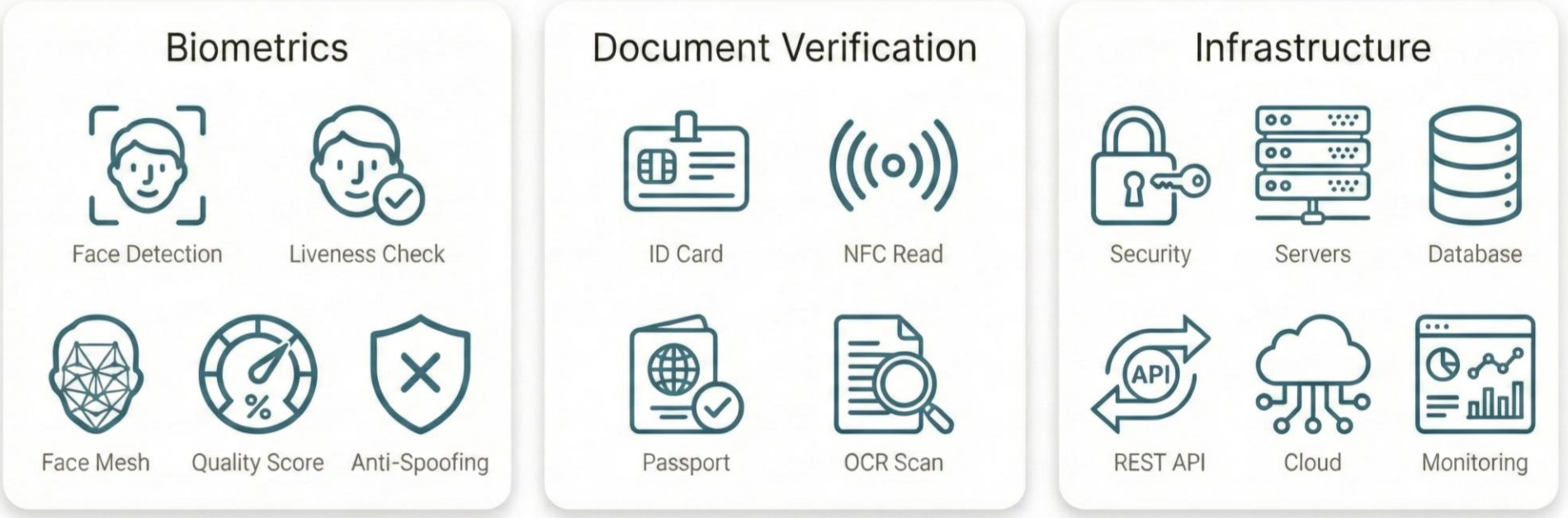


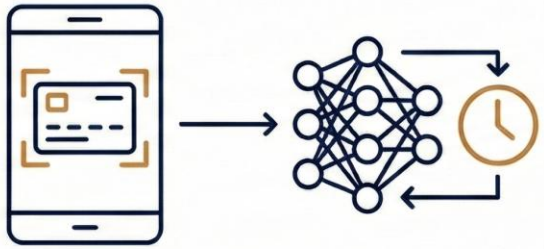
Fig. 10: Active Liveness Detection via Randomized Facial Action Challenges (a) Smile Wide Command, (b) Turn Head Left Command

TASKS ACCOMPLISHED



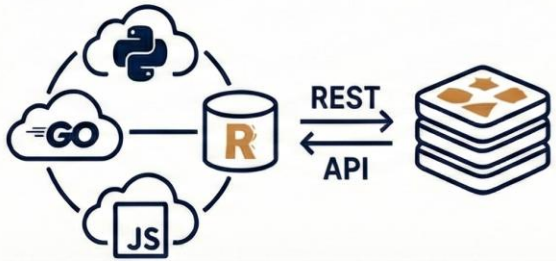
- Full-stack system developed
- Demo UI completed
- NFC reader module completed
- Robust SaaS infrastructure built
- Core biometric pipeline implemented

TECHNICAL CHALLENGES



Training a performant on-device Card Detection Model

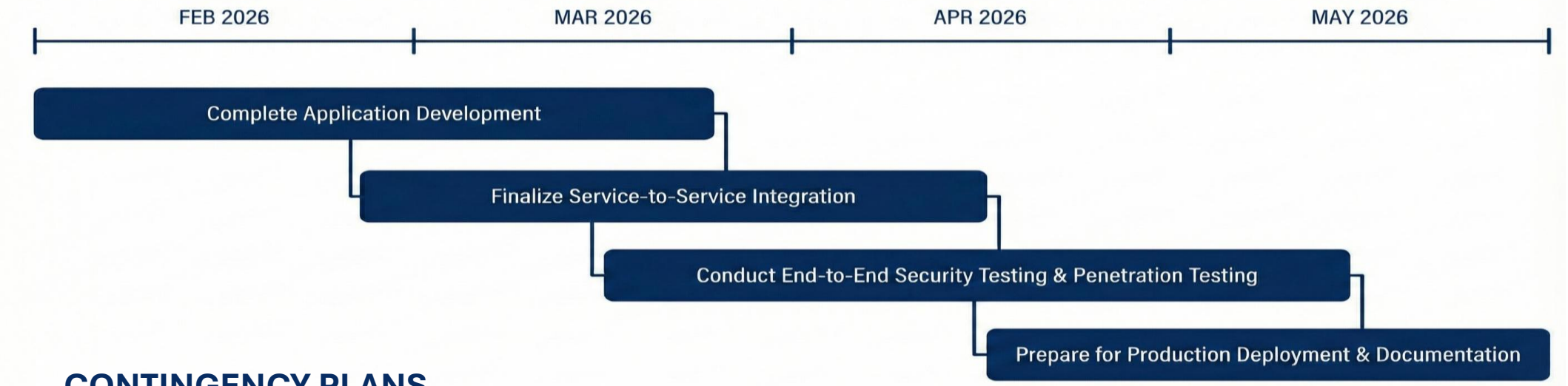
YOLO model confused Marmara student vs staff cards due to minimal visual differences in color and layout.



Cross-Language Microservice Communication

Designed a robust contract using REST APIs and a shared Redis cache for state.

SECOND SEMESTER PLAN



CONTINGENCY PLANS

NFC Failure? → Scope narrows to support Turkish eID only initially.

Mobile App Delayed? → Prioritize a Desktop-First Development strategy.

High WebSocket Latency? → Fallbacks to REST polling for status updates.

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THANK YOU & QUESTIONS